Linear Algebra Problem Sheet-1

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August 21, 2026

Abstract

The first problem sheet for Graduate Mathematical Toolkit-1 2026. Please contact the TAs for doubts at sunain.ug2023@cmi.ac.in and Avishekh@cmi.ac.in.

Let $\mathbb{F}$ is an arbitrary field; $\mathbb{F}_q$ the field with q elements. All vector spaces are over $\mathbb{F}$ unless stated otherwise. $\mathcal{L}(V,W)$ is the space of linear maps $V \to W$, $\mathcal{L}(V) = \mathcal{L}(V,V)$, $\mathcal{P}_m(\mathbb{F})$ the polynomials of degree $\leq m$, $\mathbb{F}^\infty$ the space of all sequences in $\mathbb{F}$, and $V' = \mathcal{L}(V,\mathbb{F})$ the dual space. Problems marked $\star$ are harder. Section 1 is computation, Section 2 is standard bookwork, Section 3 applies the same ideas in computer science and Section 4 are some applications of Rank Nullity. The sources of the problems are Linear Algebra done Right, Linear Algebra done wrong and few standard CS problems. Some problem suggestions were also taken from Claude.

1 Computation

Problem 1.1. *Let*

$$A = \begin{pmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 0 \\ 3 & 6 & 0 & 3 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}.$$

1. *Find the reduced echelon form of A; list the pivot columns and the free variables.*
2. *Give bases for* $\operatorname{null} A$*, the column space and the row space, and verify* $\operatorname{rank} A + \dim \operatorname{null} A = 4$.
3. *Determine exactly which $b' \in \mathbb{R}^3$ make $Ax = b'$ consistent, and write the general solution of $Ax = b$ as a particular solution plus a subspace.*

Problem 1.2. *Find A^{-1} by row reduction, where $A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$, and check your answer. Which elementary matrices did you use, and what is* $\det A$ *as read off from the reduction?*

Problem 1.3. *Let $v_1 = (1,2,0,1)$, $v_2 = (2,4,1,3)$, $v_3 = (0,0,1,1)$, $v_4 = (1,2,1,2)$ in $\mathbb{R}^4$. Decide whether the list is linearly independent, find a basis of* $\operatorname{span}(v_1,\dots,v_4)$ *consisting of some of the v_i, express the remaining v_i in that basis, and extend to a basis of $\mathbb{R}^4$.*

Problem 1.4. *Let $D\colon \mathcal{P}_3(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation. Write the matrix of D with respect to the basis $1, x, x^2, x^3$. Compute* $\operatorname{null} D$, $\operatorname{range} D$, $\operatorname{rank} D$, $\det D$*, and the smallest k with $D^k = 0$. Is D invertible? Is D injective on $\{p : p(0) = 0\}$?*

Problem 1.5. *Solve $x + 2y = 3$, $3x + y = 4$ over $\mathbb{Q}$, then over $\mathbb{F}_5$. How many solutions does each have, and what changed? Answer the same question for the system over $\mathbb{F}_7$.*

2 Bookwork

Problem 2.1. *For each set, decide whether it is a subspace, with proof or counterexample: (a) $\{f : \mathbb{R} \to \mathbb{R} \mid f(0) = 0\}$ and $\{f \mid f(0) = 1\}$ in $\mathbb{R}^{\mathbb{R}}$; (b) $\{f \in C^1(\mathbb{R}) \mid f' = f\}$ in $C^1(\mathbb{R})$ — and its dimension; (c) $\{x \in \mathbb{F}^4 \mid x_1 + 2x_2 = x_4\}$ and $\{x \in \mathbb{F}^4 \mid x_1x_2 = x_3x_4\}$; (d) $\{A \in \mathcal{M}_{n\times n}(\mathbb{F}) \mid \operatorname{trace} A = 0\}$ and $\{A \mid \det A = 0\}$.*

Problem 2.2. *Give a nonempty $U \subseteq \mathbb{R}^2$ that is (a) closed under scalar multiplication but not addition; (b) closed under addition and additive inverses but is not a subspace.*

Problem 2.3. *Let $U_1,\dots,U_m$ be subspaces of V. Prove the sum is direct iff the only way to write $0 = u_1 + \cdots + u_m$ with $u_j \in U_j$ is with every $u_j = 0$. Show that for $m \geq 3$ this is* not *implied by $U_i \cap U_j = \{0\}$ for all $i \neq j$.*

Problem 2.4. *For finite-dimensional subspaces $U, W \subseteq V$, prove* $\dim(U+W) = \dim U + \dim W - \dim(U \cap W)$. *Then show the three-subspace analogue of inclusion–exclusion fails, using three subspaces of* $\mathbb{R}^2$.

Problem 2.5. *Show that the even functions and the odd functions are subspaces of* $\mathbb{R}^{\mathbb{R}}$ *and that* $\mathbb{R}^{\mathbb{R}} = V_{\mathrm{e}} \oplus V_{\mathrm{o}}$.

Problem 2.6. *Prove* $\dim V = \dim \operatorname{null} T + \dim \operatorname{range} T$ *for* $T \in \mathcal{L}(V, W)$, *V finite-dimensional. Deduce that* $\dim V > \dim W$ *forces non-injectivity,* $\dim V < \dim W$ *forces non-surjectivity, and that a homogeneous system with more unknowns than equations has a nonzero solution.*

Problem 2.7. *Show that row rank of a matrix = column rank of a matrix*

Problem 2.8. *Define Determinant of a Matrix. Show that it is multiplicative.*

3 Applications to CS

Problem 3.1. *(Fast linear recurrences.) With* $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$, *prove* $M^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$ *and give an algorithm computing* $F_n \bmod p$ *using* $O(\log n)$ *multiplications. Generalise to any recurrence* $a_n = c_1 a_{n-1} + \cdots + c_k a_{n-k}$.

Problem 3.2. *(Linear codes.) A linear code is a subspace* $C \subseteq \mathbb{F}_q^n$; *a parity-check matrix is H with* $C = \operatorname{null} H$.

1. *Show* $\dim C = n - \operatorname{rank} H$.
2. *Prove the minimum weight of a nonzero codeword equals the smallest number of linearly dependent columns of H.*
3. *Let* $H \in \mathcal{M}_{3\times 7}(\mathbb{F}_2)$ *have as columns all seven nonzero vectors of* $\mathbb{F}_2^3$. *Compute* $\dim C$ *and the minimum distance, and explain how to correct one error by reading off* Hx.

Problem 3.3. *(Boolean functions.) Let* $\mathcal{F}$ *be the space of functions* $\mathbb{F}_2^n \to \mathbb{F}_2$. *Compute* $\dim \mathcal{F}$, *and prove the multilinear monomials* $\prod_{i \in S} x_i$ *for* $S \subseteq [n]$ *form a basis. Conclude that every Boolean function has a unique multilinear representation over* $\mathbb{F}_2$.

Problem 3.4. $\star$ *(Oddtown.) Let* $A_1, \ldots, A_m \subseteq [n]$ *with every* $|A_i|$ *odd and every* $|A_i \cap A_j|$ *even for* $i \neq j$. *Using characteristic vectors in* $\mathbb{F}_2^n$, *prove* $m \leq n$. *Where exactly does the argument use that we are working over* $\mathbb{F}_2$?

Problem 3.5. $\star$*(Fisher's inequality). Let k be a positive integer. In a town with n people, m clubs have been formed. Every two clubs share exactly k members. Prove that* $m \leq n$.

4 Applications of rank–nullity

Problem 4.1. *(Counting solutions.) Let* $A \in \mathcal{M}_{m\times n}(\mathbb{F}_q)$ *and* $b \in \mathbb{F}_q^m$. *Prove that* $Ax = b$ *has either no solution or exactly* $q^{n - \operatorname{rank} A}$ *solutions. Deduce that the number of b for which the system is solvable is* $q^{\operatorname{rank} A}$, *so a uniformly random b is solvable with probability* $q^{\operatorname{rank} A - m}$.

Problem 4.2. *Compute the following dimensions by exhibiting a surjective linear map and applying rank–nullity.*

1. $\{A \in \mathcal{M}_{n\times n}(\mathbb{F}) \mid \operatorname{trace} A = 0\}$.
2. $\{p \in \mathcal{P}_m(\mathbb{F}) \mid p(a_1) = \cdots = p(a_k) = 0\}$, *for distinct* $a_1, \dots, a_k$ *with* $k \le m+1$.
3. $\{A \in \mathcal{M}_{n\times n}(\mathbb{F}) \mid$ *every row sum and every column sum is* $0\}$. *(Careful: the natural map to* $\mathbb{F}^{2n}$ *is not surjective.)*

Problem 4.3. *Let* $A \in \mathcal{M}_{m\times n}(\mathbb{F})$ *and* $B \in \mathcal{M}_{n\times p}(\mathbb{F})$ *with* $AB = 0$. *Prove* $\operatorname{rank} A + \operatorname{rank} B \le n$, *by applying rank–nullity to* B *viewed as a map into* $\operatorname{null} A$. *When is equality attained?*

Problem 4.4. *Let* $P \in \mathcal{L}(V)$ *satisfy* $P^2 = P$. *Prove* $V = \operatorname{range} P \oplus \operatorname{null} P$ *and* $\operatorname{rank} P + \operatorname{rank}(I - P) = \dim V$. *Choosing a basis adapted to the decomposition, show* $\operatorname{rank} P = \operatorname{trace} P$ *when* $\operatorname{char} \mathbb{F} = 0$.

Problem 4.5. *(Hermite interpolation.) Fix distinct* $a_1, \dots, a_n \in \mathbb{F}$ *with* $\operatorname{char} \mathbb{F} = 0$ *and define* $T\colon \mathcal{P}_{2n-1}(\mathbb{F}) \to \mathbb{F}^{2n}$ *by* $T(p) = \big(p(a_1), p'(a_1), \dots, p(a_n), p'(a_n)\big)$. *Show* T *is linear and injective, and conclude by rank–nullity that prescribed values* and *derivatives at* n *points are met by exactly one polynomial of degree* $< 2n$.

Problem 4.6. *(No lossless linear compression.) Show that if* $m < n$ *there is no injective linear map* $\mathbb{F}^n \to \mathbb{F}^m$. *For a linear hash* $h(x) = Hx$ *with* $H \in \mathcal{M}_{m\times n}(\mathbb{F}_2)$ *and* $m < n$, *show that every* $y \in \operatorname{range} h$ *has exactly* $2^{\,n-\operatorname{rank} H}$ *preimages, and hence that collisions are guaranteed.*

Problem 4.7. $\star$ *Fix* $A \in \mathcal{M}_{n\times n}(\mathbb{F})$, $\operatorname{char} \mathbb{F} = 0$, *and let* $\Phi\colon \mathcal{M}_{n\times n}(\mathbb{F}) \to \mathcal{M}_{n\times n}(\mathbb{F})$ *be* $\Phi(X) = AX - XA$. *Show* Φ *is linear, that* $\operatorname{range} \Phi$ *lies in the trace-zero matrices, and deduce* $\dim\{X \mid AX = XA\} \ge 1$. *Improve this to* $\ge n$ *when* A *has* n *distinct eigenvalues in* $\mathbb{F}$.